Worksheet A: (Topic 2.1) Arithmetic and Geometric Sequences Name: Mrs. Camiel

Directions: For each of the following, determine if the given sequence is arithmetic, geometric, or neither.

1. $12, 7, 2, -3, -8, \ldots$

-5 -5 -5 -5

Arithmetic

Common difference: -5

2. $5, 10, 20, 40, \ldots$

$\cdot 2$ $\cdot 2$ $\cdot 2$

Geometric

Common ratio: 2

3. $20, 10, 5, \frac{5}{2}, \ldots$

$\cdot \frac{1}{2}$ $\cdot \frac{1}{2}$

Geometric

Common ratio: $\frac{1}{2}$

4. $\frac{1}{3}, 1, \frac{5}{3}, \frac{7}{3}, 3, \ldots$

$+\frac{2}{3}$ $+\frac{2}{3}$ $+\frac{2}{3}$ $+2/3$

Arithmetic

Common difference: $\frac{2}{3}$

5. $1, 1, 2, 3, 5, 8, 13, \ldots$

$+0$ $+1$ $+1$ $+2$ $+3$ $+5$

Neither.

6. $b_n \quad \frac{n+3}{2}$

$b_0 = \frac{3}{2}$ $+1/2$

$b_1 = 2$ $+1/2$

$b_2 = \frac{5}{2}$

$b_n = \frac{1}{2}n + \frac{3}{2}$

$\frac{n}{2} + \frac{3}{2}$

$\frac{3n}{2}$

$\frac{3}{2}n + 0$

Arithmetic

Common difference: $\frac{1}{2}$

Directions: Let a_n be an arithmetic sequence with the following properties. For each of the following, find an expression for a_n, and then find a_{11}.

7. $a_3 = 7$ and $a_8 = 17$ $\quad \frac{17-7}{8-3} = \frac{10}{5} = 2$

$a_n = 7 + 2(n-3)$

$a_{11} = 7 + 2(11-3)$

$= 7 + 2(8)$

$= 7 + 16$

$a_{11} = 23$

8. $a_2 = -3$ and $a_6 = -9$ $\quad \frac{-9+3}{6-2} = \frac{-6}{4} = \frac{-3}{2}$

$a_n = -3 - \frac{3}{2}(n-2)$

$a_{11} = -3 - \frac{3}{2}(11-2)$

$= -3 - \frac{3}{2}(9)$

$= -\frac{6}{2} - \frac{27}{2}$

$a_{11} = -33/2$

9. $a_5 = 7$ and $d = -4$

$a_n = 7 - 4(n-5)$

$a_{11} = 7 - 4(11-5)$

$= 7 - 4(6)$

$a_{11} = -17$

10. $a_4 = -1$ and $d = \frac{2}{3}$

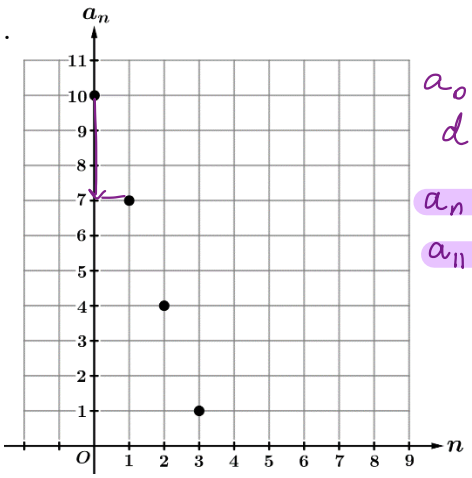
$a_n = -1 + \frac{2}{3}(n-4)$

$a_{11} = -1 + \frac{2}{3}(11-4)$

$= -1 + \frac{2}{3}(7)$

$a_{11} = \frac{11}{3}$

11.



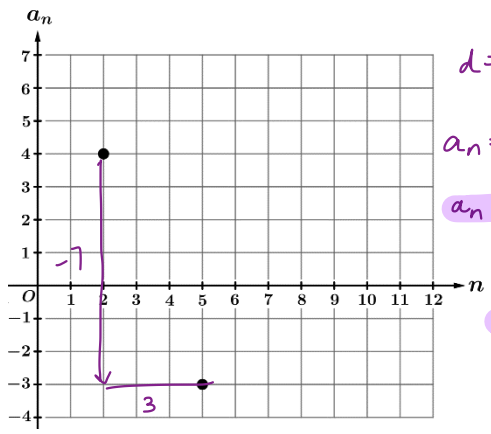
$a_0 = 10$

$d = -3$

$a_n = 10 - 3n$

$a_{11} = -23$

12.



$d = -\frac{7}{3}$

$a_n = a_2 - \frac{7}{3}(n-2)$

$a_n = 4 - \frac{7}{3}(n-2)$

$a_{11} = 4 - \frac{7}{3}(11-2)$

$a_{11} = -17$

Directions: Let g_n be a geometric sequence with the following properties. For each of the following, find an expression for g_n, and then find g_6.

13. $g_1 = 5$ and $r = -2$

$g_n = 5(-2)^{n-1}$

$g_6 = 5(-2)^{6-1}$

$= 5(-2)^5$

$g_6 = -160$

14. $g_2 = 8$ and $r = \frac{1}{2}$

$g_n = 8\left(\frac{1}{2}\right)^{n-2}$

$g_6 = 8\left(\frac{1}{2}\right)^4$

$g_6 = \frac{1}{2}$

15. $g_2 = 1$ and $g_5 = 27$

$g_n = g_2 r^{n-2}$

$g_5 = g_2 r^{5-2}$

$27 = 1 \cdot r^3$

$r = 3$

$g_n = 1 \cdot 3^{n-2}$

$g_n = 3^{n-2}$

$g_6 = 3^{6-2} = 3^4$

$g_6 = 81$

16. $g_4 = -12$ and $g_7 = \frac{32}{9}$

$g_7 = g_4 r^{n-4}$

$\frac{32}{9} = -12 \cdot r^3$

$-\frac{8}{27} = r^3$

$r = -\frac{2}{3}$

$g_n = -12 \cdot \left(-\frac{2}{3}\right)^{n-4}$

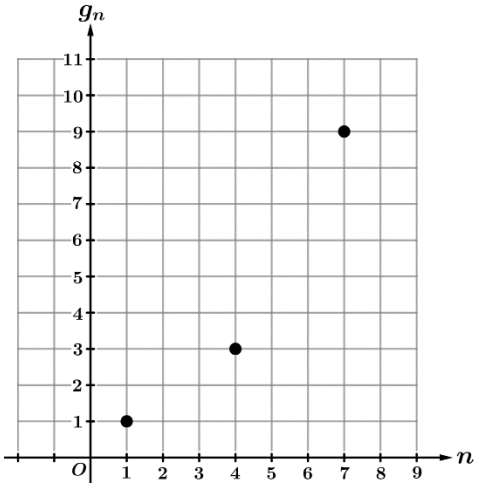
$g_6 = -12 \cdot \left(-\frac{2}{3}\right)^2$

$g_6 = -\frac{16}{3}$

$3 \div -12 \cdot \frac{4}{9 \div 3}$

$-4 \cdot \frac{4}{3}$

17.



$g_1 = 1$ $g_4 = 3$ $g_n = g_k \cdot r^{n-k}$

$g_n = g_1 \cdot r^{n-1}$

$g_4 = 1 \cdot r^{4-1}$

$3 = r^3$

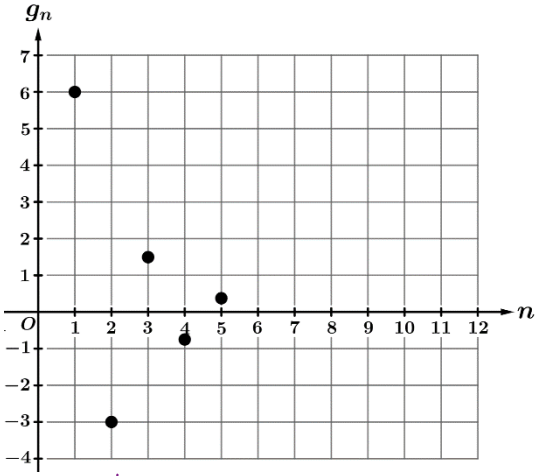
$r = \sqrt[3]{3}$

$g_n = 1 \cdot \sqrt[3]{3}^{n-1}$

$g_6 = \sqrt[3]{3}^5$

$g_6 = 3^{5/3}$

18.



$g_1 = 6$ $g_2 = -3$

$g_2 = g_1 \cdot r^{2-1}$

$-3 = 6 \cdot r^1$

$r = -\frac{1}{2}$

$g_n = 6 \cdot \left(-\frac{1}{2}\right)^{n-1}$

$g_6 = 6 \cdot \left(-\frac{1}{2}\right)^5$

$= 6 \cdot \left(-\frac{1}{32}\right)$

$g_6 = -\frac{3}{16}$